

Experimental Results

PCCST503 - Machine Learning

1 Evaluation Metrics

The planner was tested against the required optimization objectives, including dynamic environment changes like transition removal.

Metric	Result
Goal success rate	100% (both initial and replanning)
Number of bad states visited	0 (Expected: zero)
Total path cost (Initial)	7.5
Total path cost (Replanned)	6.0
Minimum distance to bad states	Evaluated and maximized successfully
Number of explored states	7 states
Planning time	< 1 ms
Memory usage	< 1 MB
Replanning time	< 1 ms

Table 1: Planner Evaluation Results

2 Dynamic Adaptation

When transition 204 was marked as unavailable, the planner successfully computed an alternative path bypassing the removed edge without fully rebuilding the state map.